

Day 3 — Fluid Mechanics II / Pipe Flow

Comprehensive Problem Set

June 19, 2026 • Concept Review + 21 Questions

Today's focus: Darcy–Weisbach friction loss, the Moody diagram & friction factor, Hazen–Williams, minor (fitting) losses, and pumps (hydraulic vs. brake power, operating point, series/parallel, NPSH).

Instructions: Work each problem with the NCEES FE Reference Handbook open. Use $g = 9.81 \text{ m/s}^2$, $\gamma_{\text{water}} = 9.81 \text{ kN/m}^3$, $\rho_{\text{water}} = 1000 \text{ kg/m}^3$, $\nu_{\text{water}} = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$ unless told otherwise. Write units at every step.

Concept Review — Pipe Flow & Pumps

1. Major (Friction) Head Loss

Darcy–Weisbach (works for any fluid, laminar or turbulent)

- $h_f = f \frac{L}{D} \frac{V^2}{2g}$ — friction head loss; note $h_f \propto V^2$ and $\propto L$, inversely $\propto D$.
- Pressure drop form: $\Delta p = \gamma h_f = f \frac{L}{D} \frac{\rho V^2}{2}$.
- Velocity/flow: $V = \frac{Q}{A}$, $A = \frac{\pi}{4} D^2$.

Friction factor f (Darcy)

- Laminar ($Re < 2100$): $f = \frac{64}{Re}$ — independent of roughness.
- Turbulent: read f from the **Moody diagram** using Re and relative roughness ε/D .
- Colebrook (implicit): $\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f}} \right)$ — *not* in the FE Handbook; on the exam read f from the Moody chart instead.
- Fully rough (complete turbulence): f depends on ε/D *only*, not Re .
- $Re = \frac{\rho V D}{\mu} = \frac{V D}{\nu}$; non-circular: hydraulic diameter $D_H = \frac{4A}{P}$.

2. Hazen–Williams (empirical; water, turbulent)

- $V = k_1 C R^{0.63} S^{0.54}$, with $k_1 = 0.849$ (SI), 1.318 (US, ft/s).
- $R = \frac{D}{4}$ (full circular pipe); $S = \frac{h_f}{L}$ (slope of energy grade line).
- C = Hazen–Williams roughness coefficient (higher C = smoother pipe; PVC ≈ 150 , new steel ≈ 120 , old cast iron ≈ 100).
- Valid for water near room temperature in turbulent flow only — *not* a general fluid relation.

3. Minor (Local) Losses

- $h_{f,\text{fitting}} = C \frac{V^2}{2g}$ — loss at fittings, valves, bends, entrances/exits. The FE Handbook uses the symbol C (not K) for the loss coefficient.

- C is given in the problem statement; typical values: sharp entrance 0.5, exit 1.0, gate valve (open) ≈ 0.2 , 90° elbow ≈ 0.9 , open globe valve ≈ 10 .
- Total head loss $h_{L,total} = \underbrace{f \frac{L}{D} \frac{V^2}{2g}}_{\text{major}} + \underbrace{\sum C \frac{V^2}{2g}}_{\text{minor}}$.

4. Energy Equation with a Pump

- $\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 + h_{pump} = \frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + h_{turbine} + h_L$.
- Pump head supplied: $h_{pump} = (z_2 - z_1) + h_L$ (lifts fluid and overcomes losses).
- **Energy line (EGL)** = total head $\frac{p}{\gamma} + \frac{V^2}{2g} + z$ above a datum; **hydraulic grade line (HGL)** = $\frac{p}{\gamma} + z$ (the EGL minus the velocity head).
- HGL and EGL fall in the flow direction by the head loss; a pump steps both lines *up* by h_{pump} , a turbine steps them *down*.

5. Pumps — Power, Curves, Affinity, NPSH

Power

- Hydraulic (water/fluid) power: $\dot{W}_{fluid} = \gamma Q H = \rho g Q H$.
- Brake (shaft) power: $\dot{W}_{brake} = \frac{\gamma Q H}{\eta}$, where η = pump efficiency.
- 1 kW = 1 kN · m/s; in US units use hp = $\frac{\gamma Q H}{550 \eta}$ (ft-lb/s).

System / operating point (conceptual — not Handbook formulas)

- System curve (concept): $H_{sys} = H_{static} + kQ^2$, losses rising with Q^2 .
- **Operating point** = intersection of the pump curve and the system curve.
- **Series** pumps: *heads add* at a given flow; **parallel** pumps: *flows add* at a given head — useful to know, but the series/parallel combination is not in the FE Handbook.

Affinity (scaling) laws & cavitation — *these are in the Handbook*

- $\frac{Q_2}{Q_1} = \frac{N_2}{N_1}$, $\frac{H_2}{H_1} = \left(\frac{N_2}{N_1}\right)^2$, $\frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3$ (speed N).
- Avoid cavitation when $NPSH_{available} > NPSH_{required}$.

Handbook: FE Reference Handbook — “Fluid Mechanics”: Hydraulic Gradient / Energy Line; Consequences of Fluid Flow (Head Loss); Moody, Darcy, or Stanton Diagram; Hazen-Williams Equation; Performance of Components (Pumps).

Questions

1. In the Darcy–Weisbach equation, friction head loss in a pipe is proportional to which power of the average velocity?
(A) V (B) V^2
(C) $1/V$ (D) V^3
2. Water flows at $V = 2.0$ m/s through a 150 mm (0.15 m) diameter pipe of length 200 m with Darcy friction factor $f = 0.02$. The friction head loss is closest to:
(A) 2.7 m (B) 5.4 m
(C) 10.9 m (D) 0.54 m
3. For fully developed laminar flow with $Re = 1800$, the Darcy friction factor is:
(A) 0.018 (B) 0.0356
(C) 0.064 (D) 0.32
4. In the fully rough (complete-turbulence) zone of the Moody diagram, the friction factor depends on:
(A) Reynolds number only (B) relative roughness ε/D only
(C) both Re and ε/D (D) neither
5. Using Hazen–Williams (SI), water flows in a 0.30 m pipe with $C = 130$ and energy slope $S = 0.005$. The velocity is closest to:
(A) 0.62 m/s (B) 1.23 m/s
(C) 2.47 m/s (D) 3.7 m/s
6. A gate valve ($C = 0.2$) carries water at $V = 2.5$ m/s. The minor head loss across the valve is closest to:
(A) 0.032 m (B) 0.064 m
(C) 0.16 m (D) 0.64 m
7. A pump delivers $Q = 0.04$ m³/s of water against a total head of 25 m. The hydraulic (water) power added to the fluid is closest to:
(A) 4.9 kW (B) 9.8 kW
(C) 14.0 kW (D) 19.6 kW
8. For the pump in the previous problem, if the pump efficiency is 70%, the required brake (shaft) power is closest to:
(A) 6.9 kW (B) 9.8 kW
(C) 14.0 kW (D) 20.0 kW

9. Two identical pumps operated in **parallel**, compared with a single pump, primarily increase the:
- (A) head delivered at a given flow
 - (B) flow delivered at a given head
 - (C) pump efficiency
 - (D) NPSH required
10. A centrifugal pump running at 1,200 rpm produces 20 m of head. Using the pump affinity (scaling) laws, if the speed is increased to 1,800 rpm, the head produced is closest to:
- (A) 30 m
 - (B) 45 m
 - (C) 67.5 m
 - (D) 13.3 m
11. Water flows at $V = 2.0$ m/s through a section with two 90° elbows ($C = 0.9$ each) and a fully open globe valve ($C = 10$). The total minor (fitting) head loss is closest to:
- (A) 1.2 m
 - (B) 2.4 m
 - (C) 0.24 m
 - (D) 4.8 m
12. Water flows at $Q = 0.10$ m³/s in a 0.25 m pipe. The average velocity is closest to:
- (A) 1.0 m/s
 - (B) 2.0 m/s
 - (C) 4.1 m/s
 - (D) 0.5 m/s
13. Water ($\nu = 1.0 \times 10^{-6}$ m²/s) flows at $V = 1.5$ m/s in a 0.10 m pipe. The Reynolds number and flow regime are:
- (A) 1.5×10^4 , laminar
 - (B) 1.5×10^5 , turbulent
 - (C) 1.5×10^5 , laminar
 - (D) 6.6×10^5 , turbulent
14. A 0.10 m pipe, 100 m long, $f = 0.02$, carries water at $V = 2.0$ m/s with a sharp entrance ($C = 0.5$) and an exit ($C = 1.0$). The total head loss (major + minor) is closest to:
- (A) 4.1 m
 - (B) 4.4 m
 - (C) 0.31 m
 - (D) 8.2 m
15. A pump lifts water a static height of 15 m while overcoming 8 m of friction loss at $Q = 0.03$ m³/s. The hydraulic power required is closest to:
- (A) 4.4 kW
 - (B) 6.8 kW
 - (C) 2.4 kW
 - (D) 9.1 kW
16. **(Numeric entry)** A pump delivers $Q = 0.05$ m³/s of water at a head of 40 m with an efficiency of 80%. Determine the required brake power, in kW. _____
17. To prevent cavitation at a pump suction, the design requirement is:
- (A) $\text{NPSH}_A < \text{NPSH}_R$
 - (B) $\text{NPSH}_A > \text{NPSH}_R$
 - (C) $\text{NPSH}_A = 0$
 - (D) $p_{\text{vapor}} > p_{\text{atm}}$

18. A friction head loss of 5.0 m of water corresponds to a pressure drop of approximately:
- (A) 5.0 kPa (B) 49 kPa
(C) 9.8 kPa (D) 490 kPa
19. A commercial steel pipe has absolute roughness $\varepsilon = 0.045$ mm and diameter $D = 100$ mm. Its relative roughness ε/D is:
- (A) 0.00045 (B) 0.0045
(C) 0.45 (D) 0.045
20. The Hazen–Williams equation is most appropriate for:
- (A) any fluid in any regime (B) water in turbulent flow
(C) laminar flow only (D) compressible gas flow
21. **(Numeric entry)** Water ($Q = 0.06$ m³/s) flows in a 0.20 m pipe, 500 m long, with $f = 0.018$. Determine the Darcy–Weisbach friction head loss, in meters. _____

Answer Key

Q	Ans	Q	Ans	Q	Ans
1	B	8	C	15	B
2	B	9	B	16	24.5 kW
3	B	10	B	17	B
4	B	11	B	18	B
5	B	12	B	19	A
6	B	13	B	20	B
7	B	14	B	21	8.4 m

Worked Solutions

1. Darcy–Weisbach $h_f = f \frac{L}{D} \frac{V^2}{2g}$ contains V^2 , so $h_f \propto V^2$. **(B)**

Handbook: Consequences of Fluid Flow — Head Loss Due to Flow (Darcy-Weisbach)

$$2. h_f = f \frac{L}{D} \frac{V^2}{2g} = 0.02 \cdot \frac{200}{0.15} \cdot \frac{(2.0)^2}{2(9.81)} = 0.02(1333)(0.2039) = 5.44 \text{ m. (B)}$$

Handbook: Consequences of Fluid Flow — Head Loss Due to Flow (Darcy-Weisbach)

$$3. \text{Laminar: } f = \frac{64}{Re} = \frac{64}{1800} = 0.0356. \text{ (B)}$$

Handbook: Moody, Darcy, or Stanton Friction Factor Diagram (laminar line $f = 64/Re$)

4. In the fully rough zone the Moody curves are horizontal: f is set by ε/D alone and is independent of Re . **(B)**

Handbook: Moody, Darcy, or Stanton Friction Factor Diagram

$$5. R = \frac{D}{4} = 0.075 \text{ m. } V = 0.849 C R^{0.63} S^{0.54} = 0.849(130)(0.075)^{0.63}(0.005)^{0.54}. (0.075)^{0.63} = 0.187, (0.005)^{0.54} = 0.0598, \text{ so } V = 110.4(0.187)(0.0598) = 1.23 \text{ m/s. (B)}$$

Handbook: Fluid Mechanics — Hazen-Williams Equation

$$6. h_{f, \text{fitting}} = C \frac{V^2}{2g} = 0.2 \cdot \frac{(2.5)^2}{2(9.81)} = 0.2(0.3186) = 0.064 \text{ m. (B)}$$

Handbook: Fluid Mechanics — Minor Losses in Pipe Fittings

$$7. \dot{W}_{fluid} = \gamma Q H = (9.81 \text{ kN/m}^3)(0.04 \text{ m}^3/\text{s})(25 \text{ m}) = 9.81 \text{ kW. (B)}$$

Handbook: Fluid Mechanics — Pump Power Equation

$$8. \dot{W}_{brake} = \frac{\gamma Q H}{\eta} = \frac{9.81}{0.70} = 14.0 \text{ kW. (C)}$$

Handbook: Fluid Mechanics — Pump Power / Efficiency

9. Pumps in parallel split the flow path, so at a common head their flows add — the combination delivers more flow. **(B)**

Handbook: Concept — series/parallel pumps are not given as Handbook equations

10. Affinity law (speed change): $H_2 = H_1 \left(\frac{N_2}{N_1} \right)^2 = 20 \left(\frac{1800}{1200} \right)^2 = 20(1.5)^2 = 45 \text{ m. (B)}$

Handbook: Fluid Mechanics — Performance of Components: Scaling/Affinity Laws

11. $\sum C = 0.9 + 0.9 + 10 = 11.8$; $h_{f,fitting} = \sum C \frac{V^2}{2g} = 11.8 \cdot \frac{(2.0)^2}{2(9.81)} = 11.8(0.2039) = 2.41 \text{ m. (B)}$

Handbook: Fluid Mechanics — Minor Losses in Pipe Fittings

12. $A = \frac{\pi}{4}(0.25)^2 = 0.0491 \text{ m}^2$; $V = \frac{Q}{A} = \frac{0.10}{0.0491} = 2.04 \text{ m/s. (B)}$

Handbook: Fluid Mechanics — Continuity ($Q = AV$)

13. $Re = \frac{VD}{\nu} = \frac{(1.5)(0.10)}{1.0 \times 10^{-6}} = 1.5 \times 10^5$. The Handbook classifies $Re > 10,000$ as fully turbulent (laminar $Re < 2,100$), so the flow is turbulent. **(B)**

Handbook: Fluid Mechanics — Reynolds Number

14. Major: $0.02 \cdot \frac{100}{0.1} \cdot \frac{2^2}{2(9.81)} = 4.08 \text{ m}$. Minor: $(0.5 + 1.0) \frac{2^2}{2(9.81)} = 1.5(0.2039) = 0.31 \text{ m}$. Total = $4.08 + 0.31 = 4.39 \text{ m. (B)}$

Handbook: Fluid Mechanics — Total Head Loss (major + minor)

15. $H = 15 + 8 = 23 \text{ m}$; $\dot{W} = \gamma QH = (9.81)(0.03)(23) = 6.77 \text{ kW. (B)}$

Handbook: Fluid Mechanics — Pump Power Equation

16. $\dot{W}_{brake} = \frac{\gamma QH}{\eta} = \frac{(9.81)(0.05)(40)}{0.80} = \frac{19.62}{0.80} = 24.5 \text{ kW. (Numeric: 24.5 kW)}$

Handbook: Fluid Mechanics — Pump Power / Efficiency

17. Cavitation is avoided when the available net positive suction head exceeds the required value: $NPSH_A > NPSH_R$. **(B)**

Handbook: Fluid Mechanics — NPSH / Cavitation

18. $\Delta p = \gamma h_f = (9.81 \text{ kN/m}^3)(5.0 \text{ m}) = 49.1 \text{ kPa. (B)}$

Handbook: Fluid Mechanics — Pressure Head ($p = \gamma h$)

19. $\frac{\varepsilon}{D} = \frac{0.045 \text{ mm}}{100 \text{ mm}} = 0.00045$. **(A)**

Handbook: Fluid Mechanics — Relative Roughness / Moody Diagram

20. Hazen–Williams is an empirical relation calibrated for water at ordinary temperatures in turbulent flow; it is not valid for general fluids, laminar flow, or gases. **(B)**

Handbook: Fluid Mechanics — Hazen–Williams Equation

21. $A = \frac{\pi}{4}(0.20)^2 = 0.0314 \text{ m}^2$, $V = \frac{0.06}{0.0314} = 1.91 \text{ m/s}$. $h_f = 0.018 \cdot \frac{500}{0.20} \cdot \frac{(1.91)^2}{2(9.81)} = 0.018(2500)(0.1859) = 8.37 \text{ m}$. (**Numeric:** 8.4 m)

Handbook: Consequences of Fluid Flow — Head Loss Due to Flow; Continuity